

Transforming Financial Leasing with Machine Learning: Intelligent Risk Assessment and Predictive Analytics

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Abstract—This research presents an integrated machine learning framework for financial leasing that improves risk estimation, operational analysis, and customer interaction. The system includes four components. First, a data-driven impairment estimation model directly predicts impairment amounts and one-year expected credit loss using historical leasing data, repayment behavior, collateral attributes, and customer financial information. Second, a branch performance analysis module evaluates operational efficiency, portfolio quality, and profitability across branches using clustering and performance scoring techniques. Third, a natural language processing based conversational chatbot provides automated assistance for leasing inquiries and internal support. Fourth, a default risk prediction model identifies customers with a high likelihood of repayment failure using classification algorithms applied to demographic, financial, and transactional features. The proposed framework supports proactive risk management, data-driven decision making, and improved service efficiency in financial leasing environments. Experimental results demonstrate that machine learning improves predictive accuracy and operational insight in the leasing domain.

Keywords—XGBoost, Probability of Default, Expected Credit Loss, Retrieval-Augmented Generation, Branch Performance Analysis

I. INTRODUCTION

The financial leasing industry has an important function in driving economic development by allowing companies to access vehicles and other assets using security-based financing. Nevertheless, leasing firms are also under great financial pressure from high levels of risk associated with customer defaults, poor performance of their branches, impaired loans, and noncompliance with applicable regulations. The increasing quantity of non-performing loans (NPLs), declining operational efficiency, and rising customer expectations for service

make it critical that new data-based approaches are developed for assessing risk and improving workflow efficiency [1].

To assist leasing companies in improving their ability to manage financial risk, machine learning (ML) and artificial intelligence (AI) are becoming increasingly valuable tools for financial risk management [2]. With their ability to process large volumes of financial and behavioral data, ML and AI can identify risk patterns that have not been previously identified and enhance credit assessment, which will allow firms to effectively take steps to manage risk [3].

This research project presents an integrated AI framework that assists the financial leasing sector in addressing many important operational and financial-related difficulties. The first component of the framework, predicting default probability, is developed using supervised ML techniques that assess the probability of default (PD) based on financial and behavioral data while providing safe data handling through FastAPI back-end processing [4]. This model supports improved credit risk management through advanced feature engineering methods by providing the ability to assign leasing customers into many different categories of credit risk [5].

The second element of the proposal outlines a Branch Performance Analysis (BPA), which leverages operational and financial data to assess the performance of branches at an individual level. By applying class-balanced machine learning techniques, the implementation allows for the identification of branches at risk for non-performing loans (NPLs) and provides management with advanced warning indications for appropriate interventions in a timely manner [6].

Also, as part of the research study described, a Retrieval-Augmented Generation (RAG) based AI assistant is built that

can help answer customer requests, using various forms of data input. This AI assistant utilizes vector-based document retrieval and natural language generation to respond with accuracy and improve customer service while reducing administrative workload [7].

In the last section of this study, the findings show how the predictive modeling framework can support IFRS 9 compliance with respect to the measurement of impairment and a 1-Year Expected Credit Loss (ECL). Specifically, it utilizes extensive feature engineering and ensemble learning to produce highly accurate predictions that will result in the early identification of any deterioration within portfolios, automate the calculation of provisions, and improve financial planning [8], [9].

In summary, the findings demonstrate that the use of ML and AI in financial leasing operations will lead to operational improvements by providing a framework that promotes intelligent, scalable, and data-driven decision-making.

II. LITERATURE REVIEW

A. Machine Learning Applications in Financial Leasing

Leasing institutions are experiencing difficulties attributed to credit risk, operational inefficiencies, adherence to regulations, and the management of customer service. Traditional decision-making systems for leases are based mainly upon rule-based systems and statistical models that are dependent on a limited number of financial indicators such as income, loan amount, and past payment history. The existing research indicates that current methods cannot accurately model the complex nature of a borrower's behavior, the non-linear relationships between variables, or the early warning signs of financial distress [1], [10].

Recently, machine learning techniques have been implemented to help overcome the limitations of existing methods [2, 3]. Logistic regression is still one of the most widely used baseline models in attempting to perform statistical analysis because it is relatively easy to interpret; however, its predictive capabilities are limited when dealing with high-dimensional and non-linear datasets [11]. Numerous ensemble learning techniques such as random forests, gradient boosting, and XGBoost have outperformed traditional machine learning methods for modeling financial risks by creating many weak learners that are combined to generate stronger models and capturing complex interactions between features [12–14]. In the academic literature, it has been reported that, compared with more traditional statistical methodologies, ensemble-based methodologies have resulted in 15–30% improvement in their ability to predict credit risk and financial leasing [15, 16].

However, the literature still suffers from a focus on isolated predictive activities rather than developing an integrated framework capable of supporting the many different financial leasing activities within a unified system.

B. Credit Risk, Default Probability, and Behavioral Analysis

Credit risk assessment is one of the most important aspects of financial leasing because it is vital to estimating any

customer's Probability of Default (PD), which is necessary for reducing Non-Performing Leases (NPLs) [17]. Historically, researchers have relied heavily on parameters such as loan size, tenor, interest rates, and demographic information to make predictions about a client's ability to repay [18].

However, studies demonstrate that models built on only static information do not provide absolute accuracy and subsequently underperform during periods of fluctuation within the economy [19].

The importance of including behavioral indicators in credit risk assessments (e.g., delayed payments, arrears ratios, loan age, and repayment consistency) has increased significantly over the past several years [20, 21]. In addition, research using Random Forest and Gradient Boosting algorithms demonstrated that by using both financial attributes and behavioral indicators together, predictive modeling provided an increased ability to predict the likelihood of a client's default [22, 23]. Some studies also indicated that using XGBoost to address class imbalance and feature interactions improves the robustness of those predictions to more than 90% accuracy [24].

Most previous studies have studied banking loan datasets instead of vehicle leasing portfolios. Therefore, they do not take into consideration the additional risks produced by asset depreciation and the unique structure of installment repayments.

C. Impairment and Expected Credit Loss Prediction under IFRS 9

The IFRS 9 accounting standard includes two new mandatory requirements: assessing credit risk and estimating expected credit loss based on forward-looking information [8]. The traditional impairment model is based on a historical average and applies fixed mathematical parameters to determine if a customer will default. The traditional model has been less successful in predicting the early stages of credit deterioration.

Academic studies have also focused on using regression-based models of impairment and ECL using ensemble methods, which employ machine learning algorithms such as Gradient Boosting, CatBoost, and LightGBM to predict non-linear relationships in financial data and effectively model skewed distributions on leasing data [9], [13]. Several high-quality feature engineering methods such as arrears ratios, payment burden indicators, interaction terms, and log transformations have successfully been used to dramatically increase the predictive accuracy of these models [14].

D. Performance Analysis and Risk Monitoring in Financial Institutions

In order to maintain the financial viability of leasing companies, it is essential to evaluate performance and identify underperforming entities. The conventional methods of evaluating performance primarily use manual analysis and static financial ratios. These forms of evaluation are very labor-intensive, typically employ a reactive approach, and often take far too long to provide results. A number of recent studies have investigated the use of machine learning algorithms

for classification to create performance-based classification hierarchies that can provide early detection of risk trends associated with entity performance [6].

Classification techniques such as Random Forest, Logistic Regression, and gradient-boosted algorithms have proven to be quite powerful with respect to working with imbalanced datasets [16]. These classification techniques have successfully identified risk patterns associated with levels of performance.

However, many of the existing machine learning solutions for monitoring performance do not have real-time prediction capabilities, automated model selection capabilities, or scalable deployment architecture. Thus, their use in fast-paced financial leasing environments is limited.

E. Intelligent Customer Interaction and AI-Based Automation

Customer service automation has become a priority in the financial services sector due to increasing customer demand and rising operational costs. Traditional rule-based chatbots offer limited flexibility and often struggle with queries requiring contextual understanding or domain-specific knowledge. Recent research shows that AI-based conversational systems using natural language processing and information retrieval can better address customer needs [19].

Retrieval-Augmented Generation (RAG) architecture can significantly improve financial service applications by combining vector embeddings and similarity search to retrieve relevant knowledge before generating responses [7]. RAG-based systems have been shown to enhance response accuracy and reliability compared to traditional language models. In addition, multimodal interaction capabilities such as Speech-to-Text (STT), Text-to-Speech (TTS), and OCR improve usability and accessibility.

Despite these advancements, limited studies have explored intelligent conversational systems specifically for financial leasing services. Existing literature highlights unresolved challenges, including integrating domain-specific knowledge, ensuring strong security protocols to protect user data, and developing systems that can scale effectively with organizational growth.

F. Research Gaps and Study Positioning

According to the literature review, there are a number of important gaps in the research on financial leasing systems. The first gap is that most of the existing studies in this area have primarily focused on single applications of machine learning rather than integrated systems that support multiple financial leasing functions. The second gap is that most credit risk and default prediction models do not adequately leverage behavioral data, and there is a lack of real-time processing capabilities with these kinds of models. The third gap is that research around predicting impairment and ECL does not use robust ensemble modeling, nor has there been significant validation on vast sources of vehicle leasing data. The fourth gap relates to performance monitoring solutions, which have largely been reactive, and there have not been scalable deployment mechanisms used for these types of

systems. Finally, there are very few instances of an AI-based customer interaction system focused on finance that is specifically designed for workflow in leasing and for engaging with customers in multiple modalities.

These gaps in the previous literature will form the basis for the proposed comprehensive machine-learning-based financial leasing system, which is designed to integrate risk prediction, impairment predictions, performance monitoring, and intelligent customer interaction within a unified, scalable, and secure architecture. The methodology section to follow will provide an overview of the system design, data processing methods, and model implementation approaches that will be employed to overcome the limitations described in the previous literature review.

III. METHODOLOGY

This section presents the systematic approach adopted to develop an integrated machine learning framework for financial leasing risk management, encompassing system architecture, data preprocessing, feature engineering, model development, and deployment.

A. System Architecture and Framework Design

The proposed system implements a microservices architecture consisting of four interconnected modules: Default Risk Prediction, Impairment and Expected Credit Loss (ECL) Estimation, Branch Performance Analysis, and an AI-powered Conversational Assistant, designed to ensure scalability, modularity, and real-time processing. Backend services are developed using FastAPI with Python 3.9+, with models serialized via joblib and deployed as RESTful API endpoints validated through Pydantic models. Supabase provides the primary PostgreSQL database infrastructure with row-level security and schema normalized to Third Normal Form. The conversational assistant utilizes FAISS for vector storage supporting the RAG architecture, while the frontend implemented in Next.js with React presents interactive dashboards via RESTful API calls.

B. Data Collection and Preprocessing

1) *Dataset Description:* The research utilizes 532,482 vehicle leasing records from a Sri Lankan financial institution (2025–2026), spanning 30 primary features across customer demographics, financial attributes, behavioral patterns, collateral information, and transaction records. A subset of 188,866 records was used for default risk model training and evaluation. The dataset includes branch information, financial attributes (facility amount, tenor, rates, net rental, down payment), payment behavior, arrears details, asset information, and derived metrics such as the arrears intensity ratio.

2) *Data Cleaning and Quality Assurance:* Missing values in numerical features were handled using median imputation for features with less than 5% missingness; features exceeding 40% missing values were excluded. Missingness in the dataset was minimal (FlatRate: 0.16%, Equipment Type: 0.60%, Last Receipt Paid Amount: 0.62%). Categorical missing values were assigned an 'Unknown' category, and derived feature missing values were imputed with zero.

3) *Feature Engineering and Transformation*: The binary default label assigns 1 to ‘Write Off’ accounts and 0 otherwise. Three engineered features support the default risk model: the Loan-to-Value Ratio (facility amount relative to down payment), the Payment Coverage Ratio (net rental relative to periodic repayment), and the Arrears Severity score (composite of arrears intensity and rentals in arrears). For impairment and ECL prediction, eight derived features are constructed. Log transformations are applied to the facility amount and net rental to handle skewed distributions, as defined in (1):

$$x_{\log} = \log(1 + x) \quad (1)$$

where x represents the original feature value (facility amount or net rental). This transformation compresses the large dynamic range of monetary values, reducing the influence of extreme outliers on model training.

Polynomial features (Tenor², Age², Arrears², Effective Rate², Effective Rate³) are also added to capture non-linear relationships. A composite risk score quantifies borrower exposure by combining the effective interest rate with the number of rentals in arrears, as defined in (2):

$$\text{Risk Score} = \frac{\text{No. of Rentals in Arrears} \times \text{Effective Rate}}{100} \quad (2)$$

Higher values of this score indicate simultaneous pressure from both repayment delinquency and financing cost, making it a strong early-warning signal for default and impairment estimation.

The Arrears Ratio, used as the primary classification criterion for branch performance labelling, is computed as follows:

$$\text{ArrearsRatio} = \frac{\text{ArrearsCapital} + \text{ArrearsInterest}}{\text{FacilityAmount} + 1} \quad (3)$$

A small constant of 1 is added to the denominator to prevent division by zero in edge cases. Records exceeding defined thresholds on this ratio, rental arrears count, NPL status, payment recency, and capital arrears severity are labelled ‘Poor’; otherwise they are labelled ‘Good’. Label encoding was applied to categorical features with target encoding for high-cardinality variables. All numerical features were standardized using StandardScaler fitted on the training set to prevent data leakage.

C. Default Risk Prediction Model Development

Four algorithms were evaluated: Logistic Regression (L2 regularization baseline), Decision Tree (non-linear boundaries with interpretability), Random Forest (bootstrap aggregation ensemble), and XGBoost (selected as primary model for superior performance with built-in regularization and class imbalance handling via `scale_pos_weight`). The 188,866-record dataset was partitioned 80/20 using stratified sampling (90.2% non-default, 9.8% default). Grid Search with 5-fold cross-validation over 162 XGBoost combinations yielded the optimal configuration: `n_estimators=100`,

`max_depth=3`, `learning_rate=0.01`, `subsample=0.7`, `colsample_bytree=0.7`. For the Branch Performance model, undersampling produced a balanced dataset of 78,849 samples (52,566 Good, 26,283 Poor).

Fig. 1 illustrates the predicted default probabilities across the test set. The left panel demonstrates clear separation between non-default and default classes across sample indices, while the right panel confirms that newer accounts (lower age values) carry substantially higher default risk, with red tones indicating actual defaults concentrated at low age and high predicted probability.

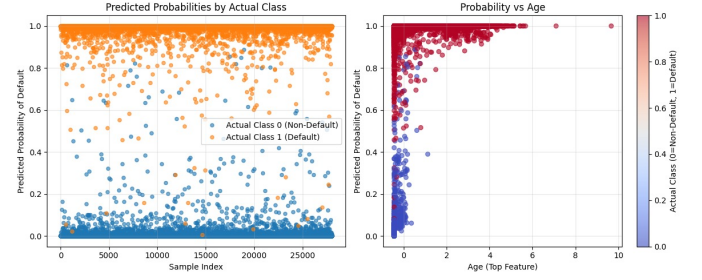


Fig. 1. Predicted probabilities by actual class (left) and probability versus account age (right). Newer accounts show markedly higher default risk.

Fig. 2 presents the Decision Tree confusion matrix. While 26,234 non-default cases were correctly classified, 11,281 actual defaults were misclassified as non-default, revealing a high false-negative rate and motivating the adoption of XGBoost as the primary model.

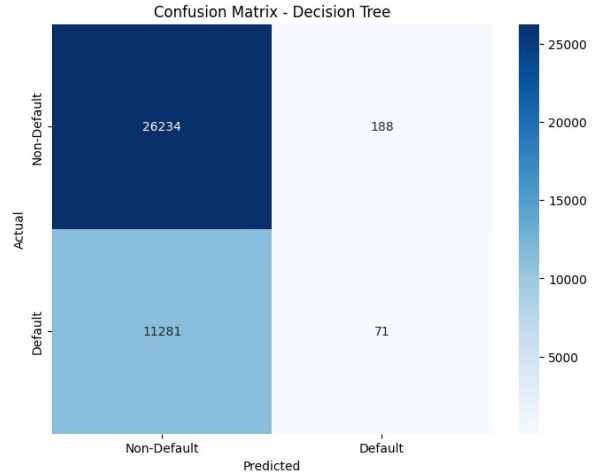


Fig. 2. Confusion matrix for the Decision Tree baseline. The 11,281 false negatives highlight poor minority class recall under class imbalance.

Fig. 3 ranks the top 10 features by importance. Granted Date and Last Receipt Paid Amount are the strongest predictors, followed by Facility Amount and Net Rental, confirming that temporal and repayment behavior features are the primary drivers of default risk.

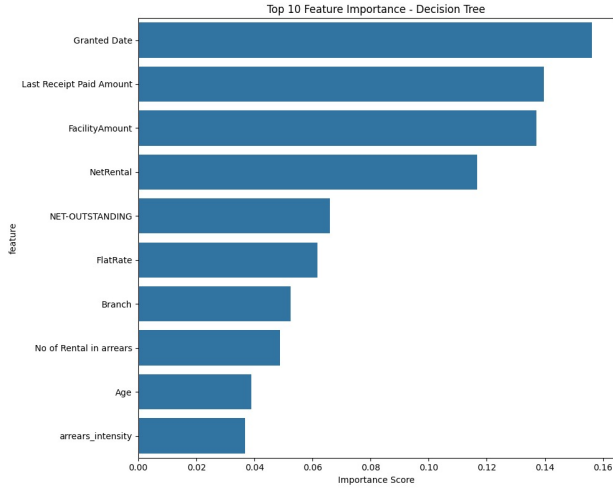


Fig. 3. Top 10 Decision Tree feature importances. Granted Date and Last Receipt Paid Amount dominate, indicating that account recency and payment activity are critical default signals.

D. Impairment and Expected Credit Loss Prediction

ECL estimation follows IFRS 9 accounting standards, predicting Impairment amount and 1-Year ECL. Six regression algorithms were implemented: Random Forest (200 estimators, depth 20), XGBoost (200 estimators, depth 10, lr 0.1), LightGBM (200 estimators, depth 15, 50 leaves), CatBoost (200 iterations, depth 10, lr 0.1), Gradient Boosting (200 estimators, depth 10), and Ridge Regression with Polynomial Features ($\alpha = 1.0$). The 99,888-record dataset was split 80/20. CatBoost achieved the best R^2 for Impairment (0.9886) and 1-Year ECL (0.9446) and was selected for deployment.

E. Branch Performance Analysis System

Branch performance is evaluated using binary classification ('Good'/'Poor') based on the Arrears Ratio defined in (3), rental arrears count, NPL status, payment recency, and capital arrears severity. Four classifiers were evaluated on the balanced 78,849-sample dataset split 80/20, with Arrears Ratio, Rentals in Arrears, and NPL Status identified as the strongest predictors.

F. RAG-based Conversational AI Assistant

The assistant implements RAG combining vector-based retrieval with natural language generation for text and voice interactions. The knowledge base covering leasing policies, product specifications, and regulatory guidelines is chunked into 512-token segments with 50-token overlap. Chunks are embedded into 384-dimensional vectors via sentence-transformers and stored in FAISS. At inference, the top-5 retrieved chunks above a 0.7 cosine similarity threshold are combined with the last 5 conversation turns to generate grounded responses, minimizing hallucination. Multimodal support includes Speech-to-Text (STT), Text-to-Speech (TTS), and direct text input through the web interface.

G. Model Deployment and Evaluation

Models are serialized via joblib and exposed through FastAPI RESTful endpoints with Pydantic validation, returning JSON responses with probability scores and risk classifications. Classification models are evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC; the XGBoost default model achieved 99.98% Accuracy, 100.00% Precision, 99.78% Recall, 99.89% F1-Score, and 100.00% ROC-AUC. Regression models use MAE, RMSE, R^2 , and MAPE. All models underwent 5-fold stratified cross-validation. The implementation stack includes Python 3.9+, scikit-learn 1.3+, XGBoost 2.0+, LightGBM 4.0+, CatBoost 1.2+, FastAPI 0.104+, Supabase (PostgreSQL 15+), FAISS 1.7+, sentence-transformers, Next.js 14+, and React 18+.

IV. LIMITATIONS AND FUTURE WORK

The proposed framework has several limitations. The dataset originates from a single Sri Lankan financial institution covering vehicle leasing records, which may limit generalizability to other regions or asset categories. Class imbalance in the default dataset (90.2% vs 9.8%) may introduce bias despite mitigation strategies, and the impairment models may not fully capture sudden macroeconomic shocks. The RAG assistant is constrained by its static knowledge base, and the system has not been evaluated under large-scale real-time production conditions.

Future work should focus on expanding the dataset across multiple institutions and regions, and incorporating dynamic macroeconomic indicators to enhance default and ECL predictions. Federated learning could enable cross-institutional model training while preserving privacy. The conversational assistant could be improved with multilingual support and RLHF. Deeper integration of explainability tools such as SHAP and LIME would support regulatory compliance, and evaluation under real-time streaming conditions using Apache Kafka would validate production-grade scalability.

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